

A lacunary counterexample to the reverse weak difference inequality

New counterexample for the question in arXiv:2102.09657

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Counterexample theorem. Fix $N \geq 1$, $1 < p < \infty$, and $0 < s < 1$. There is no constant C such that every

$$u \in \mathcal{F}^{-1}[C_c^\infty(\mathbb{R}^N \setminus \{0\})]$$

satisfies

$$\|u\|_{\dot{F}_{p,2}^s} \leq C \left[\frac{u(x) - u(y)}{|x - y|^{N/p+s}} \right]_{L^{p,\infty}(\mathbb{R}^N \times \mathbb{R}^N)}. \quad (1)$$

Indeed, there are u_m for which the norm on the left is $\gtrsim \sqrt{m}$, while the weak norm on the right stays bounded. Consequently the holomorphic-family formulation (1.29) in the source is also false.

The source question

Theorem 1.3 of Brezis–Seeger–Van Schaftingen–Yung proves the forward bound

$$\left[\frac{u(x) - u(y)}{|x - y|^{N/p+s}} \right]_{L^{p,\infty}} \lesssim \|u\|_{\dot{F}_{p,2}^s}.$$

The paragraph after it asks whether this can be reversed. Using $\dot{F}_{p,2}^s = [L^p, \dot{W}^{1,p}]_s$, it equivalently asks for a holomorphic interpolation family whose boundary norms are bounded by the weak difference quotient.

then to control the left hand side of (1.28), we only need to pay a price of a smaller constant $C''(N, p)[s(1-s)]^{-1/p}$ (as opposed to $C''(N, p)[s(1-s)]^{-1/2}$). This follows from an adaptation of the arguments given for Proposition 1.4 which we will not give in detail.

An interesting related question is whether the inequality in (1.27) can be reversed. Since $\dot{F}_{p,2}^s(\mathbb{R}^N) = [L^p(\mathbb{R}^N), \dot{W}^{1,p}(\mathbb{R}^N)]_s$, this question could be reformulated as follows: If $N \in \mathbb{N}$, $p \in (1, \infty)$, $s \in (0, 1)$ and $u \in \mathcal{F}^{-1}[C_c^\infty(\mathbb{R}^N \setminus \{0\})]$, is there a holomorphic family of functions $\{u_z(x) : z \in \mathbb{C}, 0 \leq \operatorname{Re} z \leq 1\}$ so that $u_s(x) = u(x)$, and so that

$$\max \left\{ \sup_{\operatorname{Re} z=0} \|u_z\|_{L^p(\mathbb{R}^N)}, \sup_{\operatorname{Re} z=1} \|\nabla u_z\|_{L^p(\mathbb{R}^N)} \right\} \lesssim \left[\frac{u(x) - u(y)}{|x - y|^{\frac{N}{p}+s}} \right]_{L^{p,\infty}(\mathbb{R}^N \times \mathbb{R}^N)} ? \quad (1.29)$$

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Source evidence: arXiv:2102.09657, PDF page 7 (unaltered crop).

Construction

Choose a power $R = 2^K \geq 8$ and a nonzero Schwartz function ϕ such that $\widehat{\phi} \in C_c^\infty(B(0, \varepsilon))$, with $\varepsilon > 0$ small. For $m \geq 1$, put

$$u_m(x) = \phi(x)P_m(x_1), \quad P_m(t) = \sum_{j=1}^m R^{-js} e^{2\pi i R^j t}. \quad (2)$$

The Fourier support of u_m is contained in the finite disjoint union

$$\bigcup_{j=1}^m B(R^j e_1, \varepsilon),$$

which is compact and avoids the origin. Thus every u_m belongs to the source's test class.

The Triebel–Lizorkin norm diverges

Use an equivalent smooth dyadic resolution adapted to the widely separated frequency balls. Its projection at level Kj equals the identity on $B(R^j e_1, \varepsilon)$ and annihilates the other selected balls. Hence

$$\Delta_{Kj} u_m = R^{-js} \phi(x) e^{2\pi i R^j x_1}.$$

Keeping only these m terms in the square function gives

$$\begin{aligned} \|u_m\|_{\dot{F}_{p,2}^s} &\gtrsim \left\| \left(\sum_{j=1}^m \left| 2^{Kjs} R^{-js} \phi(x) e^{2\pi i R^j x_1} \right|^2 \right)^{1/2} \right\|_{L_x^p} \\ &= \sqrt{m} \|\phi\|_{L^p}, \end{aligned}$$

because $2^K = R$. Equivalence of admissible Littlewood–Paley resolutions has constants independent of the finite number m of bands.

A uniform Hölder bound

The elementary geometric-series estimate

$$\sum_{j=1}^m R^{-js} \min\{2, 2\pi R^j r\} \leq A_{R,s} r^s \quad (r > 0) \quad (3)$$

holds uniformly in m . To see it, split where $R^j r$ crosses 1. Before that scale, sum the geometric progression $r R^{j(1-s)}$; after it, sum R^{-js} . Both pieces are bounded by a constant times r^s . It follows that

$$|P_m(a) - P_m(b)| \leq A_{R,s} |a - b|^s, \quad |P_m(a)| \leq (R^s - 1)^{-1}, \quad (4)$$

uniformly in m .

Fix $M > N/p$ and let $w(x) = (1 + |x|)^{-M}$. Schwartz decay and the mean-value theorem imply

$$|\phi(x)| \leq B_M w(x), \quad |\phi(x) - \phi(y)| \leq B_M |x - y|^s (w(x) + w(y)). \quad (5)$$

For the second bound, use the gradient estimate when $|x - y| \leq 1$ (points on the segment have comparable weights) and the zeroth-order estimate when $|x - y| > 1$.

Writing $r = |x - y|$ and expanding the product in (2), equations (4)–(5) yield

$$\begin{aligned} |u_m(x) - u_m(y)| &\leq |\phi(x) - \phi(y)| |P_m(x_1)| + |\phi(y)| |P_m(x_1) - P_m(y_1)| \\ &\leq C_{\phi,R,s,M} r^s (w(x) + w(y)), \end{aligned} \quad (6)$$

again with no dependence on m .

The weak quotient stays bounded

Divide (6) by $r^{N/p+s}$. It remains to estimate

$$H(x, y) = \frac{w(x) + w(y)}{|x - y|^{N/p}}.$$

This weak estimate is explicit. For every $\lambda > 0$,

$$\begin{aligned} \mathcal{L}^{2N} \left\{ (x, y) : \frac{w(x)}{|x - y|^{N/p}} > \lambda \right\} &= \int_{\mathbb{R}^N} \left| B \left(x, (w(x)/\lambda)^{p/N} \right) \right| dx \\ &= \kappa_N \lambda^{-p} \|w\|_{L^p}^p. \end{aligned}$$

Applying this once to $w(x)$ and once to $w(y)$, with threshold $\lambda/2$, gives

$$[H]_{L^{p,\infty}(\mathbb{R}^{2N})} \leq 2(2\kappa_N)^{1/p} \|w\|_{L^p} < \infty.$$

Therefore (6) proves

$$\sup_{m \geq 1} \left[\frac{u_m(x) - u_m(y)}{|x - y|^{N/p+s}} \right]_{L^{p,\infty}(\mathbb{R}^{2N})} < \infty. \quad (7)$$

Together with the $\sqrt{m}$ lower bound, (7) contradicts every putative constant in (1).

Why the holomorphic-family version also fails

If the family requested in source display (1.29) existed with a uniform implicit constant, the definition of complex interpolation would give

$$\|u\|_{[L^p, \dot{W}^{1,p}]_s} \lesssim \left[\frac{u(x) - u(y)}{|x - y|^{N/p+s}} \right]_{L^{p,\infty}}.$$

On the source's frequency-localized class, $[L^p, \dot{W}^{1,p}]_s = \dot{F}_{p,2}^s$ with equivalent norms. Taking $u = u_m$ contradicts (7). Thus the reformulation fails along with the direct reverse inequality.

Proof intuition

Each frequency band is scaled so that it contributes one unit to the Littlewood–Paley square function; m orthogonal scales therefore accumulate as $\sqrt{m}$. The weak quotient behaves differently. At a separation $|x - y| \simeq R^{-k}$, the k th frequency is the largest relevant term; lower and higher frequencies form convergent geometric tails. Taking a weak supremum sees the worst scale but does not add the m scale contributions. This square-function versus supremum mismatch is exactly the obstruction to reversal.

Scope and novelty. The counterexample works for every fixed $N \geq 1$, $1 < p < \infty$, and $0 < s < 1$, using complex-valued Schwartz functions with compact Fourier support away from zero. It disproves the unrestricted reverse inequality and the stated holomorphic-family bound. Bounded exact-phrase and concept searches through 13 August 2026 found no later resolution of this exact question; human literature review is still recommended.

Human review. Check the adapted Littlewood–Paley projection on the separated balls, the two geometric sums in (3), and the distribution calculation for H . These are the only substantive estimates.

References

- [1] H. Brezis, A. Seeger, J. Van Schaftingen, and P.-L. Yung, *A new formula for the L^p norm*, arXiv:2102.09657 (2021), question following Theorem 1.3 on PDF page 7.